Physics Guess Paper Class 12 Federal Board, 2026

Chapter # 15: Gravitation

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions

- Orbital Velocity – Derivation
- Geostationary Satellites – Derivation
- Gravitational Potential – Derivation

Short Questions

3) Why does an apple fall towards the Earth due to gravity, while the Earth doesn't move towards the apple?

4) Can gravitational field strength be negative? Explain.

Numerical: Examples are Mandatory

3) What will be value of 'g' on an exoplanet (a planet outside the Solar System) whose mass is five times the mass of Earth and its radius is twice the radius of Earth? **(Ans: 12.25 $m\,s^{-2}$)**

5) To launch a satellite around planet Mars at an altitude of 300 km above its surface. Mars has a mass of 6.42×10^{23} kg and has a radius of 3.39×10^{6} m, calculate its orbital velocity.

(Ans: 3.4 $km\,s^{-1}$)

6) An asteroid orbits the sun 8.35×10^{11} m from it. (a) How fast must the asteroid travel to maintain its circular orbit around the sun? (b) How long will it take the asteroid to orbit the sun? **(Ans: 1.26×10^{4} $m\,s^{-1}$, 13.2 years)**

7) Mars has a mass of 6.42×10^{23} kg and has a period of 88,642 s. What would be the radius of a stationary satellite for this planet? **(Ans: 20,428 km)**

8) Calculate the potential energy of the Moon having mass 7.35×10^{22} kg, relative to Earth with mass 6×10^{24} kg if the distance between their centres is 3.94×10^{5} km. **(Ans: -7.4×10^{28} J)**

10) What is the change in gravitational potential energy of a 64.5 kg astronaut, lifted from Earth's surface into a circular orbit of altitude 4.40×10^{2} km? **(Ans: 2.6×10^{8} J)**

Chapter # 16: Statistical Mechanics and Thermodynamics

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

PRESSURE EXERTED BY GAS MOLECULES

$$P = \frac{mN}{3V}\langle v^2 \rangle \qquad \text{——— (16.10)}$$

Temperature in terms of Average Translational Kinetic Energy of Gas Molecules

$$T \propto \langle \frac{1}{2}mv^2 \rangle \qquad \text{or} \qquad T \propto \langle K.E \rangle$$

16.2 ROOT MEAN SQUARE SPEED OF AN IDEAL GAS

Short Questions

6) Show that the ratio of the root mean square speeds of molecules of two different gases at a certain temperature is equal to the square root of the inverse ratio of their masses.

Numerical: Examples are Mandatory

1) The mass of a helium atom is 6.64×10^{-27} kg. Calculate the root mean square speed of helium atom in a gas at a temperature of 15 °C. (Ans: 1.34×10^{3} m s^{-1})

3) The mass of a molecule of a gas is 6.4×10^{-27} kg. Calculate the root mean square speed of gas molecule and kinetic energy per molecule at temperature 400 K.

(Ans: 1.6×10^{3} m s^{-1}, 8.3×10^{-21} J)

4) At which temperature the root mean square speed of gas molecules becomes double than its speed at 27 °C, (pressure of gas is kept constant)? (Ans: 927 °C)

5) Determine the root mean square speed of argon atoms at temperature 40 °C. The molar mass of argon is 39.95 g mol^{-1}. (Ans: 442 m s^{-1})

6) Calculate the number of gas molecules in a cubic meter of gas at standard temperature and pressure (STP). (Ans: 2.68×10^{25} molecules)

Chapter # 17: Simple Harmonic Motion

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

17.7 ENERGY CONSERVATION IN SHM

$$T.E = \frac{1}{2}m\omega^2 x_o^2 \quad (17.16)$$

17.6 SIMPLE PENDULUM

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Explain Terms: displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations.

Damped Osciallations and its Types

Short Questions

Free and Forced Oscillations

1) If we halve the length of a simple pendulum to its original length, what is the alteration in the period of this pendulum? What is its new frequency?
2) If the amplitude of vibration of a body executing SHM is doubled, what will happen to the maximum kinetic energy?
3) When marching soldiers are about to cross a bridge, they break steps. Why?

Resonance

Numerical: Examples are Mandatory

1) The amplitude of the motion of a mass attached to a spring is 2.48 m, while maximum speed of its mass is 4.36 m s^{-1}. What is the period of the motion? **(Ans: 3.57 s)**

4) A mass of 8.0 kg is attached to a spring and oscillates with amplitude of 0.25 m and has a frequency of 0.60 Hz. What is the energy of the motion? (Ans: 3.6 J)

5) Calculate the period and frequency of a 3.5 m long pendulum at the following locations:

a) at Karachi, where g = 9.832 m s^{-2}.

b) at K-2, where g = 9.782 m s^{-2}.

(Ans: (a) 3.747 s, 0.2669 Hz (b) 3.757 s, 0.2662 Hz)

6) In a car engine, a piston executes SHM with amplitude of 0.41 m. The engine is running at angular frequency of 2400 rpm (251 rad s^{-1}). What is the maximum speed of the piston?

(Ans: 103 m s^{-1})

Chapter # 18: Diffraction and Interference

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

Young's Double Slit Experiment

Diffraction Grating

Short Questions

1) How is an interference pattern formed by a diffraction grating different from the pattern formed by a double slit?
2) A beam of light always spreads out. Why can a beam not be produced with parallel rays to prevent spreading?
3) In the sunlight, the shadow of a building has fuzzy edges even if the building does not. Is this a refraction effect? Explain.
4) A laser pointer emits a coherent beam of parallel light rays. Does the light from such a source spread out at all? Explain.
5) A beam of light passes through a single slit to create a diffraction pattern. How will the spacing of the bands in the pattern change if the width of the slit is increased?
7) Suppose a monochromatic light falls on a diffraction grating. What happens to the interference pattern if the same light falls on a grating that has more lines per centimeter.
8) What is the significance of the equation $d\sin(\theta) = m\lambda$ in the context of (a) diffraction gratings (b) Young's double-slit experiment?
9) What is the effect of following aspects of diffraction grating on the resulting interference pattern: (a) number of slits (b) width of the slits?

Numerical: Examples are Mandatory

2) Find the distance from the central maximum to the third minimum on the screen, if the light of wavelength 400 nm is used. The distance between the slits and screen is 0.5 m and slit separation is 5 μm? **(Ans: 0.024 mm)**

3) The distance between the slits and screen is 1m, fringe spacing on screen is 1 mm. Find the slit separation when light wavelength is 500 nm. **(Ans: 0.05 mm)**

4) Find the minimum angle for non-zero diffracted light, having slit spacing 2.0 μm with 400 nm light wavelength. (Ans: $\theta_{min} = 11.54°$ (minimum angle for first minimum))

5) The deviation of second order diffracted image formed by an optical grating having 5000 lines per centimeter is 32^0. Calculate the wavelength of light used. **(Ans: 5.3×10^{-5} cm)**

Chapter # 19: Electric Potential and Capacitor

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

19.2 ELECTRIC POTENTIAL DUE TO A POINT CHARGE

$$V = \frac{1}{4\pi\varepsilon_o}\left(\frac{Q}{r}\right)$$

Capacitance of a Parallel Plate Capacitor

19.4 COMBINATIONS OF CAPACITORS

19.5 ENERGY STORED IN A CAPACITOR

Short Questions

2) If you wish to store a large amount of energy in a capacitor bank, would you connect capacitors in series or parallel? Explain.

4) What is the net amount of charge on a charged capacitor?

5) Write some applications of capacitors in real life.

6) Would you place the plates of a parallel-plate capacitor closer together or farther apart to increase their capacitance?

Numerical: Examples are Mandatory

1) A heart defibrillator delivers 4×10^2 J of energy by discharging a capacitor initially at 1×10^4 V. What is its capacitance?

(Ans: 8 μF)

2) Two capacitors of capacitance $C_1 = 6\ \mu F$ and $C_2 = 3\ \mu F$ are connected in series. Calculate the equivalent capacitance.

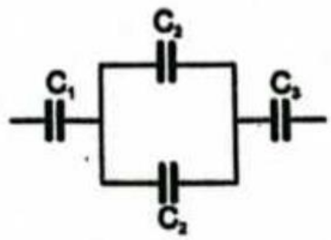
(Ans: 2μF)

3) A $2200 \mu F$ capacitor is charged up with a 1.5 V cell. Calculate the charge and energy stored in the capacitor.

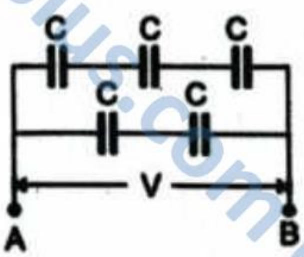
(Ans: 3.3 mC, 0.00248 J)

4) Find the equivalent capacitance for the following circuit if C_1 = 1 pF, C_2 = 0.5 pF, C_3 = 1 pF.

(Ans: 1/3 pF)



6) A network of five capacitors of capacitance C is connected to a 100 V supply, as shown in below figure. Determine the equivalent capacitance of the network. (Ans: 5C/6)



Chapter # 20: Alternating Current

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

20.2 RECTIFICATION

20.4.2 A.C Through an Inductor

20.4.2 A.C Through a Capacitor

20.5 IMPEDANCE

RL Series AC Circuit RC Series AC Circuit

Short Questions

Learn the terms Cycle, Time Period, Frequency , Average value, Peak value, RMS value, Mean & Max. Power to alternating current or voltage

1) Define (a) mutual-induction (b) self-induction.
2) Does the SI unit used for mutual-inductance and self-inductance are same? Explain briefly.

8) Distinguish between root-mean-square (r.m.s.) and peak values for a sinusoidal alternating current.

10) What is the difference between reactance and impedance?

11) Why does a capacitor block DC but allow AC to pass through?

12) How does the reactance of a capacitor change with an increase in frequency?

13) What is the phase difference between voltage and current in a (a) capacitor (b) Inductor?

14) How does the capacitance of a capacitor affect the current through it in an AC circuit?

Numerical: Examples are Mandatory

1) What is the rms value of the voltage of an ac supply having peak voltage 300 V? (Ans: 212 V)

2) An inductor with an inductance of 100 μH passes a current of 10 mA when its terminal voltage is 6.3 V. Calculate the frequency of A.C supply. (Ans: 10^6 Hz)

3) A coil of pure inductance 318 mH is connected in series with a pure resistance of 75 Ω. The voltage across resistor is 150 V and the frequency of power supply is 50 Hz. Calculate impedance of the circuit. (Ans: 124.9 Ω)

4) A resistor of resistance 30 Ω is connected in series with a capacitor of capacitance 79.5 μF across a power supply of 50 Hz and 100 V. Find impedance of the circuit. (Ans: 50 Ω)

Chapter # 21: Quantum Physics

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Theoretical + Derivations)

21.2 PHOTOELECTRIC EFFECT

21.3 COMPTON'S EFFECT

21.4 PAIR PRODUCTION

21.5 PAIR ANNIHILATION

Short Questions

1) Prove that energy and momentum of Photon are directly proportional to each other.
2) Convert 800 MeV into joules.
3) Write the two phenomena to describe photon-electron interaction.
4) In the interpretation of the photoelectric effect, how is it known that an electron does not absorb more than one photon?
5) How you can determine the work function from a plot of the stopping potential versus the frequency of the incident radiation in a photoelectric effect experiment? Can you determine the value of Planck's constant from this plot?
6) Speculate how increasing the temperature of a photo electrode affects the outcomes of the photoelectric effect experiment.
7) Which aspects of the photoelectric effect cannot be explained by classical physics?
8) What is the physical significance of Compton's effect?
9) Differentiate between Compton's shift and Compton's wavelength.
10) How can you say that scattering angle of photon plays effective role in measurement of Compton's shift?
11) What are the conditions required for pair production to occur?
12) What is meant by de Broglie wavelength?
13) Which phenomenon provides evidence for particulate nature of electromagnetic radiation?

Numerical: Examples are Mandatory

1) In an experiment, the work function of potassium surface is found to be 2.1 eV. What should be the wavelength of incident radiation if the stopping potential for the electron is 0.43 V?

(Ans: 5.91×10^{-7} m)

2) Light of wavelength 5000 Å falls on a metal surface having work function of 1.9 eV. Find (a) Energy of photon in electron-volt. (b) Kinetic energy of photo-electrons emitted. (c) Stopping potential. **(Ans: 2.48 eV, 0.58 eV, 0.58 V)**

3) Calculate the energies of the photons associated with the (i) violet light of 413 nm. (ii) X-rays of 0.1 nm. (iii) radio waves of 10 m. **(Ans: 3eV; 12424eV; 1.24×10^{-7} eV)**

4) Calculate the minimum energy required by a photon to transfer half of its energy to an electron at rest. **(Ans: 0.25 MeV)**

5) An electron is placed in a box about the size of an atom that is about 1 Å. What is the velocity of the electron? **(Ans: 7.29×10^{6} m s^{-1})**

6) A 50 keV X-ray is scattered through an angle of 90°. What is the energy of the X-ray after Compton scattering? **(Ans: 45.5 keV)**

Chapter # 22: Nuclear Physics

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

Short Questions

Numerical: Examples are Mandatory

Chapter # 23: Cosmology

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Derivations)

22.3 RADIOACTIVITY

22.4 HALF LIFE AND RATE OF DECAY

22.5.1 Nuclear Fusion

22.5.3 Nuclear Fission

22.7 ENERGY IN ANNIHILATION REACTIONS

Short Questions

Define and use the terms mass defect and binding energy.

1) What are some of the potential benefits and drawbacks of using nuclear energy as a source of electricity compared to other forms of energy?

6) If U-238 undergoes alpha decay, what is the resulting nucleus?
7) How do chain reactions occur in nuclear fission?

Numerical: Examples are Mandatory

1) If an isotope has a half-life of 10 days and there are initially 5000 nuclei, how many will remain after 30 days? (Ans: 625)

2) If a nucleus of carbon-14 has a decay constant of 0.00012 s^{-1}, what is its half-life? (Ans: 5775 s)

3) What is energy released when a nucleus of U-235 undergoes fission and release two neutrons? (Ans: 1.793×10^{-11} J)

4) Calculate the number of nuclei of C-14 remain un-decay after 40110 years if the initial atoms were 10,000 (The half-life of C-14 is approximately 5,730 years). (Ans: 4670)

5) Calculate the mass defect and binding energy per nucleon for silver (Ag) nucleus if it's atomic mass in 107.905949 u. (Ans: 41.031×10^{-12} J/nucleon)

6) Calculate energy released in the following reaction:

$${}^{1}_{0}n + {}^{238}_{92}U \rightarrow {}^{96}_{38}Sr + {}^{140}_{54}Xe + 3\,{}^{1}_{0}n + Q$$

(Ans: -6.474×10^{-11} J)

Chapter # 24: Earth's Climate

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Theory + Derivations)

23.1 BLACK BODY RADIATION

23.2 LUMINOSITY AND RADIANT FLUX INTENSITY

Short Questions

3) What is the inverse square law for radiant flux intensity?

8) How redshift leads to the idea that the Universe is expanding?

10) How do different types of spectra (emission, absorption, continuous) provide insights into the composition and properties of celestial objects?

Numerical: Examples are Mandatory

1) There are two stars, let named A and B. Which star has the greater temperature? Given that: $L_A = L_B = 10^4$ J/s; $R_A = 10^4$ m; $R_B = 10^5$ m. (Ans: star A)

2) The Sun has a surface temperature of 6000K produces peak radiation of 420 nm. Find out the temperature of the Sirius if peak radiation of Sirius is 72 nm. (Ans: 35000 K)

3) If a source has a luminosity of 3.826×10^{26} W and is at a distance of 3 lightyear, what is the radiant flux intensity? (Ans: 3.78×10^{-8} W m^{-2})

Chapter # 25: Medical Imaging

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Theoretical)

24.1 EARTH'S CLIMATE SYSTEM

24.2 OCEAN CURRENTS AND WINDS

24.3 CLIMATE INERTIA

Short Questions

1) Describe Earth's climate system as a complex system having five interacting components.
2) What is climate inertia? Explain.
3) Explain how global climate is determined by energy transfer from the Sun.

4) Discuss the Earth energy budget in detail.
5) Explain how the energy imbalance between the poles and the equator can affect atmospheric circulation.

7) Discuss that how differences in density of ocean water play an important role in ocean circulation.

Numerical: Examples are Mandatory

1) If the volume of change in storage of the hydrosphere, whose area is 30 km² is 600,000 m³, find net precipitation (P) if the volume of evapotranspiration is 1,500,000 m³ and runoff volume is 600,000 m³. (Ans: 90 mm)

2) If a Coriolis force of 30 N deflects a mass of air at 30⁰ moving with speed of 40 km h⁻¹. Find the mass of this air packet (where $\omega = 7.3 \times 10^{-5}$ rad s^{-1}). (Ans: 369,900 kg)

Chapter # 26: Nature of Science: A Debate

MCQs -> Exercise MCQs + Model + Past Papers of 2025 (Mandatory)

Long Questions (Theoretical)

25.1 PIEZOELECTRIC EFFECT AND ULTRASONIC WAVES

25.2 X-RAYS

Short Questions

1) Differentiate between piezoelectric effect and inverse piezoelectric effect.
2) What is the distinction between a continuous X-ray and a characteristic X-ray?
3) What is the main purpose of X-rays?
4) X-rays can emit electrons from metal surface and X-rays can be diffracted. Comment.
5) Why X-rays have different properties from light even though both originate from orbital transition of electrons in excited atoms?
6) What is meant by breaking radiation?
7) What are K_β characteristic X-rays? Show with the help of diagram.

Numerical: Examples are Mandatory

1) If potential difference of 3000 V is applied across an X-ray tube. What will be the minimum wavelength of X-rays produced, if the electrons were slowed down by the target. (Ans: 0.414 nm)

2) Find the voltage across the X-ray tube through which an electron must be accelerates in order to generate the continuous X-ray of exactly 0.1 nm. (Ans: 12, 400 V)